[0x25]

bit cycling

ie f_{rcx} : .loop$_{start}$ → .loop$_{end}$ = id_{rcx}

```
xor     eax,eax
mov     rcx,1
shl     rcx,0x20
.loop:
movzx   rbx,cx
imul    rbx,rbx

ror     rcx,0x10
movzx   rdx,cx
imul    rdx,rdx
rol     rcx,0x10

add     rbx,rdx
shr     rbx,0x20
cmp     rbx,1
adc     rax,0
loop    .loop
```

values stay the same?

shows how ror/rol/shr cycling can work

YES!! a pure cycle ie Id ≠ 0

FOR rcx

ie f_{rcx} : .loop$_{start}$ → .loop$_{end}$ = id

$g(rcx) = rol(ror(rcx, 0x10), 0x10) = rcx$

(not entire loop obv...)

[0x25]

ie the proj. $\pi_{rcx}(f(state)) = state_{rcx}$ ie $\pi_{rcx} \circ f = id$